Digital PSAT/NMSQT Math — Formula Cheat Sheet

Same 4 domains as the SAT · 44 questions · scored 160–760 · certpathpublishing.store

📐 ALGEBRA

Lines & Linear Equations

$$y = mx + b$$

- **Slope:** $m = (y_2 - y_1)/(x_2 - x_1)$
- **Point-slope:** $y - y_1 = m(x - x_1)$
- **Standard:** $Ax + By = C$ → slope = $-A/B$
- **Parallel:** equal slopes · **Perp:** $m_1 \cdot m_2 = -1$

Systems & Inequalities

- Solve 2×2 by **substitution** or **elimination**.
- **No solution:** same slope, different intercept (parallel).
- **Infinite solutions:** identical lines.
- Flip the inequality sign when ×/÷ by a negative.

🔢 ADVANCED MATH

Quadratics

$$x = [\ -b \pm \sqrt{(b^2 - 4ac)}\ ] / 2a$$

- **Vertex form:** $y = a(x - h)^2 + k$ → vertex (h, k)
- **Axis / vertex x:** $x = -b/2a$
- **Discriminant $b^2 - 4ac$:** >0 two real · =0 one · <0 none
- **Sum of roots** = $-b/a$ · **Product** = c/a

Exponents & Functions

- $x^a \cdot x^b = x^{a+b}$ · $x^a/x^b = x^{a-b}$ · $(x^a)^b = x^{ab}$
- $x^0 = 1$ · $x^{-n} = 1/x^n$ · $x^{1/n}$ = the n-th root of x
- **Growth / decay:** $y = a(1 \pm r)^t$ (r = rate per period)
- **Composition** $f(g(x))$: evaluate the inner function first.

📊 PROBLEM-SOLVING & DATA

Percent, Ratio, Rate

- **Percent change:** (new – old)/old × 100
- **Percent of:** part = percent (as decimal) × whole
- **After x% off:** price = original × (1 – x)
- **Proportion:** $a/b = c/d$ → cross-multiply

Statistics & Probability

- **Mean** = sum ÷ count · **Median** = middle value
- Outliers pull the **mean**; the **median** resists them.
- **P(event)** = favorable ÷ total outcomes
- Bigger random sample → smaller margin of error ($\propto 1/\sqrt{n}$).

📏 GEOMETRY & TRIGONOMETRY

Triangles & Circles

- **Pythagorean:** $a^2 + b^2 = c^2$
- **45-45-90:** $x : x : x\sqrt{2}$ · **30-60-90:** $x : x\sqrt{3} : 2x$
- **Circle:** $A = \pi r^2$ · $C = 2\pi r$
- **Circle equation:** $(x - h)^2 + (y - k)^2 = r^2$
- Similar triangles: areas scale by (ratio)².

Trig — SOHCAHTOA

$$\sin = O/H \quad \cos = A/H \quad \tan = O/A$$

- **Cofunction:** $\sin\theta = \cos(90° - \theta)$
- Common triples: 3-4-5, 5-12-13, 8-15-17.
- Angles in a triangle sum to 180°.

🥇 **National Merit:** For juniors, your math score feeds the **Selection Index = (2 × Reading/Writing + Math) ÷ 10** (range 48–228). The PSAT/NMSQT omits the SAT's hardest items and has less trig, so **top accuracy on easy–medium questions** matters most. 🖥️ **Desmos & Bluebook:** graph to find intercepts/vertices; type-in answers take a fraction (5/2) or decimal — never a mixed number. Geometry/volume formulas are provided on screen; **memorize** the quadratic formula, slope, and exponent rules.